A subcubic quantum Fourier transform for the symmetric group

Nate Stemen

IonQ

23 October 2026

Research draft

Abstract

We construct a uniform approximate quantum Fourier transform over the symmetric group S_n using $n^{5/2}\,\mathrm{poly}(\log(n+1), \log(1/\epsilon))$ elementary gates and $n\,\mathrm{poly}(\log(n+1), \log(1/\epsilon))$ live qubits. The construction implements the standard induction transform by preparing its columns, coherently erasing a transversal label, and applying oblivious amplitude amplification. At level M, the column circuit and its complete source and target reflections each have near-linear cost. The erased block has normalization $\sqrt{M}$, so amplification gives a $\widetilde{O}(M^{3/2})$ induction step. The implementation uses bounded fixed-point products for branching probabilities and a reversibly maintained tableau cache for selected representation matrices. All shape-dependent arithmetic, finite-gate synthesis, boundary updates, and workspace cleanup are included. For $\Lambda = O(\log(n+1) + \log(1/\epsilon))$, explicit sufficient bounds are $O(n^{5/2}\Lambda^5)$ Clifford+T gates and $O(n\Lambda^3)$ live qubits. The output channel has diamond-norm error at most ϵ on the valid input subspace. Small-instance circuit checks support the implementation contracts but do not establish the asymptotic result or a practical advantage.

1 Introduction

The Fourier transform over S_n changes from a basis of permutations to a basis indexed by irreducible representations and their matrix entries. Its ambient dimension is $n!$, but an efficient quantum algorithm acts on only polynomially many qubits. The algorithmic question is therefore not whether the Fourier matrix can be written down. It is whether its structure can be implemented with a small, uniformly generated circuit.

Beals established a polynomial-time quantum Fourier transform over the symmetric group [1]. The subgroup-chain framework was subsequently developed by Moore, Rockmore, and Russell [2], and explicit symmetric-group circuits were studied by Kawano and Sekigawa [3, 4]. Bruinsma, Grinko, and Ozols give a detailed elementary-gate analysis and a $\widetilde{O}(n^3)$ implementation with $\widetilde{O}(n^{3/2})$ qubits [5, Theorem 1.1]. We quote their theorem's stated space bound; the workspace discussion in their proof suggests a potentially tighter bound, so we do not claim a strict improvement over the best space achievable by that construction. Their analysis also illustrates why a count of abstract representation rotations does not by itself settle the complexity: the coefficients, controls, encodings, and work registers must be implemented coherently. Here and below, soft-O notation suppresses polynomial factors in $\log(n+1)$ and $\log(1/\epsilon)$.

We implement the full induction transform in a different way. A near-linear circuit prepares the required column while retaining its transversal label. A change of basis on that label exposes the

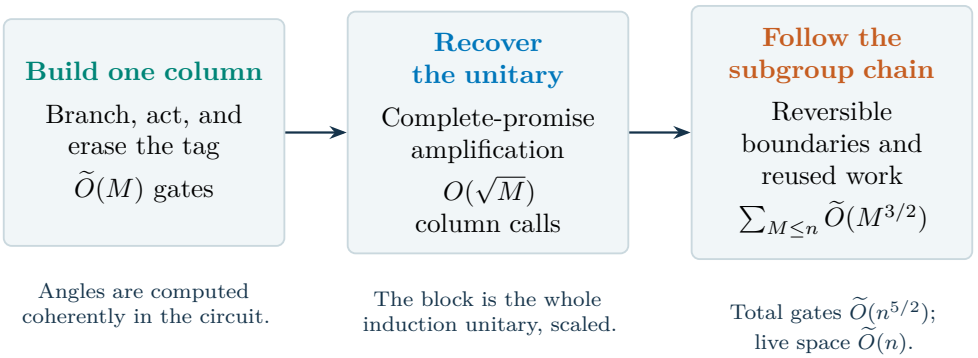


Figure 1: The argument in three scales. The first two boxes implement one induction level. The final box sums levels, not separate copies of their workspace. No measurement or postselection occurs.

entire induction unitary as a block of amplitude $1/\sqrt{M}$. Oblivious amplitude amplification converts this block into a clean induction step using $O(\sqrt{M})$ column calls. Thus

$$\underbrace{\widetilde{O}(M)}_{\text{column and promise checks}} \quad \underbrace{O(\sqrt{M})}_{\text{amplification}} = \widetilde{O}(M^{3/2}), \qquad \sum_{M=2}^{n} \widetilde{O}(M^{3/2}) = \widetilde{O}(n^{5/2}). \tag{1}$$

The polynomial gate bound improves by a factor $\sqrt{n}$ relative to the cited cubic full-QFT bound. This is an asymptotic comparison, not a claim about constants at accessible sizes.

What makes the cost bound coherent. The shape λ is a quantum register, not a classical input available to a preprocessing algorithm. Its branching probabilities and rotation angles are computed reversibly. A fixed sorting network builds the within-row ranks needed by Young's representation matrices. The cache is updated when tableau entries move, then erased from the final tableau. The amplification reflections check the *complete* source and target promises, including all column workspace. Finally, a clean outer boundary updates the representation label and the second tableau. These implementation obligations account for the gate and space bounds in the theorem.

Contributions and attribution. The induction identity, Young's orthogonal rule, dimension-ratio branching states, and amplitude amplification are established ingredients. Our contribution is a uniform near-linear coherent implementation of the column circuit and promise reflections, yielding a $\widetilde{O}(M^{3/2})$ full induction step. The proof includes a bounded-product precision argument, fixed-address tableau bookkeeping, a source-isometry error bound for two-promise amplification, and a global workspace-reuse argument. It does not require a fast decomposition of the local interlacing A-matrix, and it makes no claim about the optimal exact Givens count of that matrix.

1.1 Relation to previous constructions

The subgroup chain $S_1 \subset S_2 \subset \cdots \subset S_n$ and its multiplicity-free branching rule organize both the established algorithms and this construction [1, 2, 5, 6]. In the Beals-style implementation developed by Bruinsma, Grinko, and Ozols, successor-state preparation and controlled transversal representations already appear [5, Section 4 and Appendix A.2]. Their implementation assembles induction by processing transversal elements sequentially. We instead retain the transversal index, erase it in a uniform basis, and amplify the resulting unitary block. Sorting-network corner extraction also appears in their Appendix A.3.1. We do not claim these individual primitives as new.

Construction	Transform	Gates	Live qubits
Bruinsma–Grinko–Ozols [5, Thm. 1.1]	Full QFT	$\widetilde{O}(n^3)$	$\widetilde{O}(n^{3/2})$
Rathore–Zhou [11, Thm. 5.1]	Character transform	$\widetilde{O}(n^{5/2})$	$O(n)$
This work, Theorem 1	Full QFT	$\widetilde{O}(n^{5/2})$	$\widetilde{O}(n)$

Table 1: Scope matters as well as the exponent. The first and last rows concern the full Fourier transform. The middle row concerns a different, smaller transform and is included to delimit the claim, not as a like-for-like runtime comparison. The first row reproduces a stated theorem bound, not a space lower bound. The character-transform estimate is for inverse-polynomial error. Prior rows count one-/two-qubit gates; our row includes finite-gate synthesis.

The amplification step uses the oblivious-amplification principle of Berry et al. [7, Lemma 3.6]. Our formulation isolates the two different promise subspaces and makes clear why source-isometry approximation suffices: the image reflection depends only on the prepared source subspace, not on an arbitrary completion of its unitary. Finite synthesis is also standard [8, 9]. Theorem 4.5 of van Melkebeek and Watson supplies the finite-precision *bit-model* uniformity needed here [9].

A separate line concerns the quantum character transform (QCT). Rathore and Zhou obtain a $\widetilde{O}(n^{5/2})$-gate, $O(n)$-qubit QCT using Hamiltonian methods [11, Theorem 5.1]. The QCT, also studied in [12], acts on conjugacy-class and irreducible-character labels. It is a different task from the full $n!$-dimensional QFT, whose output includes *both* representation-matrix indices. Consequently, a matching exponent for the QCT does not establish the full-QFT bound proved here. Conversely, generic preparation, erasure, or amplification ideas should not be attributed exclusively to this work.

1.2 How to read the proof

Figure 1 gives the cost argument before its implementation details. Section 2 fixes the encodings and error model. Section 3 introduces the induction map through a worked four-box example. Sections 4–6 construct a near-linear column circuit. Section 7 explains the coherent amplification, and Section 8 assembles the full transform. Readers interested in bit complexity can then inspect the explicit precision ledger in Section 9. The final sections separate the analytic guarantee from the prototype and state the remaining implementation limitations.

2 The transform, its encodings, and its error model

The group S_n consists of all permutations of n objects, with composition as its product. A representation assigns to each permutation g a matrix $R(g)$ so that $R(gh) = R(g)R(h)$. An irreducible representation has no nontrivial invariant subspace. For S_n, these basic matrix representations are indexed by partitions of n. The Fourier transform reorganizes a function, or a quantum amplitude vector, into their matrix coefficients.

A partition $\lambda \vdash m$ is a nonincreasing list of positive integers with sum m. Its boxes have coordinates (r, c), where $1 \le c \le \lambda_r$, and content $c - r$. A standard Young tableau fills the boxes by $1, \dots, m$ in increasing order along every row and column. Write $d_\lambda = |\operatorname{SYT}(\lambda)|$. We encode a tableau by its row word $P = (p_1, \dots, p_m)$: label j is in row p_j. Its column is the within-row rank

$$c_j = |\{h \le j : p_h = p_j\}|.$$

The endpoint shape is stored as a zero-padded list of row lengths, not in a succinct or random-access data structure. Every word has $O(\log(n+1))$ bits.

Let R_λ denote Young's real orthogonal representation, with positive off-diagonal entries for adjacent transpositions. With the rightmost permutation acting first, our Fourier convention is

$$\mathcal{F}_n|g\rangle = \sum_{\lambda\vdash n} \sum_{Q,R\in \mathrm{SYT}(\lambda)} \sqrt{\frac{d_\lambda}{n!}}\, \langle Q|R_\lambda(g)|R\rangle\, |\lambda, Q, R\rangle. \tag{2}$$

The shape register is redundant but is retained. Inputs use the unique transversal word

$$g = t_{i_n}^{(n)} \cdots t_{i_1}^{(1)}, \qquad t_i^{(M)} = (i, i+1, \ldots, M), \quad 1 \le i_M \le M.$$

The valid input and output subspaces have dimension $n!$. A circuit has an arbitrary unitary extension on other physical bitstrings.

Linear-algebra conventions. A ket $|x\rangle$ denotes a basis vector, and a superposition is a linear combination of such vectors. A unitary preserves all inner products. An isometry I satisfies $I^\dagger I = \mathrm{I}$: it embeds a smaller logical space into a larger physical space, for example by appending zero work qubits. Its image projector is $II^\dagger$. A reflection $2II^\dagger - \mathrm{I}$ acts as $+1$ on that image and -1 on its orthogonal complement. The operator norm $\|A\| = \sup_{\|\psi\|=1}\|A\psi\|$ measures worst-case vector error. The channel diamond norm additionally allows entanglement with an arbitrary untouched reference system.

We count gates from the fixed finite Clifford+T basis, including reversible arithmetic and rotation synthesis. There is no QRAM, precomputed shape oracle, free real-number arithmetic, or postselection. The public parameters are n and an integer requested precision. All loops, capacities, and wire addresses are fixed by these parameters. A clean-error bound means operator norm on the valid input subspace with all auxiliary registers zero, comparing against the ideal output with auxiliaries zero. It therefore includes leakage into auxiliary registers and holds with an arbitrary entangled reference system. Approximate cleanup means this norm bound, not an assertion that every synthesized circuit ends with exactly zero auxiliary bits.

Theorem 1 (Main bound)**.** *For $n \ge 1$ and $0 < \epsilon \le 1/2$ there is a uniform finite-gate circuit with clean-input isometry $\widetilde{V}_n$ such that*

$$\|\widetilde{V}_n - (\mathcal{F}_n \otimes |0\rangle_{\mathrm{work}})\| \le \epsilon/2.$$

Its elementary-gate count and maximum number of live qubits are respectively

$$n^{5/2}\,\mathrm{poly}(\log(n+1), \log(1/\epsilon)), \qquad n\,\mathrm{poly}(\log(n+1), \log(1/\epsilon)).$$

After discarding work, the channel is within diamond distance ϵ of the ideal channel on the input promise. The same dominant bound holds for a one-line permutation input after a reversible polynomial-logarithmic overhead, quadratic-time encoding conversion.

The $\widetilde{O}$ notation below suppresses polynomial factors in $\log(n+1)$ and $\log(1/\epsilon)$; it is not uniform in unbounded exact-real precision. The result is an approximation theorem, not an exact finite-gate realization. The proof optimizes the power of n, not the logarithmic exponents or practical constants.

What coherence requires. For a classical shape, one could calculate an angle and then choose a rotation. Here the input may contain $\sum_\lambda a_\lambda|\lambda\rangle$. Measuring λ would destroy its relative phases. The implementation must instead compute $|\lambda\rangle|0\rangle \mapsto |\lambda\rangle|\widetilde{\theta}_\lambda\rangle$, apply the bit-controlled rotation, and erase the angle register. The theorem counts all three operations. The classical compiler sees only n and the requested error, never the quantum shape.

The channel's domain. Let $\mathcal{H}_{\text{in}}$ be the abstract space with orthonormal basis $\{|g\rangle : g \in S_n\}$, embedded into the valid transversal encodings. Both channels in the theorem have domain $\mathcal{L}(\mathcal{H}_{\text{in}})$. The diamond-norm comparison includes arbitrary reference systems on this domain. No approximation to a prescribed map is claimed on invalid physical bitstrings.

3 The induction transform: one box at a time

Fix $\lambda \vdash M-1$. The notation $\mu \succ \lambda$ means that μ is obtained by adding one box to λ. If $P \in \mathrm{SYT}(\lambda)$, write $P \to \mu$ for the tableau obtained by placing M in that box. Define

$$(F_\lambda)_{(\mu,Q),(i,P)} = \sqrt{\frac{d_\mu}{M d_\lambda}}\, \langle Q|R_\mu(t_i^{(M)})|P \to \mu\rangle. \tag{3}$$

The standard multiplicity-free induction rule says that this is a unitary map from $\mathbb{C}^M \otimes \mathbb{C}^{d_\lambda}$ to $\bigoplus_{\mu\succ\lambda} \mathbb{C}^{d_\mu}$. In particular, $\sum_{\mu\succ\lambda} d_\mu = M d_\lambda$. These representation-theoretic facts, and the resulting QFT recurrence, are established ingredients, not claims of this paper; see Definition 3.6, Lemma 3.7, and Theorem 3.8 of [5]. We prove the recurrence's circuit and error costs below.

For clarity, the induction identity can be understood by decomposing $\mathrm{Ind}_{S_{M-1}}^{S_M} R_\lambda$. In the coset basis it has labels (i, P). Each successor representation occurs once, its restricted λ sector has basis $P \to \mu$, and the matrix coefficients of the intertwiner are (3). Orthogonality fixes the normalization. This is the only non-elementary representation-theoretic input to the induction map itself.

3.1 A four-box example

Take $M = 4$ and $\lambda = (2, 1)$. There are two standard tableaux of shape λ. Its three successors are $(3, 1)$, $(2, 2)$, and $(2, 1, 1)$, with dimensions $3, 2, 3$. The branching probabilities are consequently $3/8, 1/4, 3/8$. For a particular tableau P, the preparation B appends the label 4 in the three possible boxes, with the square roots of these probabilities as amplitudes (Figure 2). The displayed tableau has row word $(1, 1, 2)$ and column cache $(1, 2, 1)$. Its successors append row labels 1, 2, and 3, respectively. This eight-dimensional fixed-λ induction block is one part of the construction, not the full 24-dimensional Fourier transform over S_4.

This preparation is only the first part of induction. The tag $i \in \{1, 2, 3, 4\}$ selects the permutation t_i, whose representation acts on the enlarged tableau. Erasing the tag in a uniform basis then produces the matrix block $F_\lambda/2$. A single successful block would have probability $1/4$ on any normalized input. The algorithm does not measure that block. It amplifies it coherently, preserving superpositions of tags, tableaux, and shapes.

The distinction between a prepared column and a unitary is worth keeping in view. Preparing the right output for one classical input does not by itself implement the linear transformation on an unknown superposition. Retaining the tag makes different tag sectors orthogonal; within each sector the column map is an isometry. The clean map therefore extends to a unitary. Coherent tag erasure and the exact unitary block identity are what make amplification possible.

3.2 A common physical register space

At a fixed stage M, the unchanged external register stores λ in M padded row-length slots. There are a tag register with alphabet $\{0, \ldots, M\}$, M tableau row slots, an attenuation flag, and work

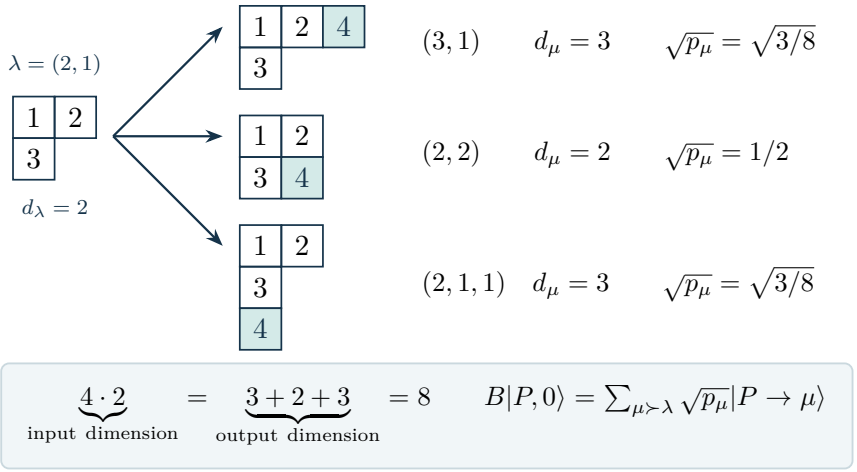


Figure 2: Branching at the first three-successor shape. A highlighted box contains the new label, not a measured choice. The preparation is coherent over the three successors and over the two possible input tableaux. The probabilities are dimension ratios, not a uniform distribution over successor shapes.

registers. The tag value 0 and row value 0 are reserved as blanks. Define isometries I_λ, J_λ into this same physical space:

$$I_\lambda|i, P\rangle = |\lambda; i, P, 0; 0; 0_W\rangle, \quad 1 \le i \le M,$$
$$J_\lambda|\mu, Q\rangle = |\lambda; 0, Q; 0; 0_W\rangle, \quad \mu \succ \lambda.$$

Here 0_W includes *all* work used by the column unitary, including coefficient, angle, column-cache, and sorting registers. The external shape is part of the control, not an erased output. Define I, J, F by direct sum over valid λ. Their ranks vary by shape, but their physical wire counts do not. Right tableaux and unused future input tags are spectators; they are not tested for zero. Inactive tails of larger-capacity stage registers are also untouched spectators, initially zero.

Two reflections will be required on these *full* promises:

$$R_I = 2II^\dagger - \mathrm{I}, \qquad R_J = 2JJ^\dagger - \mathrm{I}. \tag{4}$$

Testing only the tag is not sufficient. Dedicated predicate helper registers are kept separate from W-work and return exactly to zero after each reflection. All identities involving these reflections are on the invariant subspace where those dedicated helpers start at zero.

Register	Role during a column call	Promise-reflection treatment
Shape λ	Coherent control; unchanged	Valid size-$(M-1)$ partition
Tag and left tableau	Input (i, P); output $(0, Q)$	Full source or target validity
Attenuation flag	Sets the known block amplitude	Must be zero
Column work W	Arithmetic, cache, sorting history	Every active work bit must be zero
Predicate helpers	Used only by the reflections	Start clean; uncomputed exactly
Right tableau, future tags	Untouched spectators	Not tested for zero

Table 2: Which registers belong to the promise? The distinction prevents two errors: reflecting only on a zero tag, and accidentally rejecting nonzero spectator data.

4 Near-linear reversible tableau bookkeeping

Lemma 2 (Fixed-schedule word operations)**.** *For row words and padded shapes with M slots, the following functions have clean reversible XOR implementations using $M\,\mathrm{poly}(\log(n+1))$ gates and work bits: within-row column ranks, endpoint row histograms, standard-tableau validity, partition validity, and the condition $\mu \succ \lambda$. Consequently the reflections (4), including zero tests of W-work of size $M\,\mathrm{poly}(\log(n+1), \log(1/\eta))$, have cost and helper space $M\,\mathrm{poly}(\log(n+1), \log(1/\eta))$.*

Proof. Use a fixed bitonic sorting network on a power-of-two padding of $O(M)$ records. Its $O(M \log^2(M+1))$ compare-exchanges compare $O(\log(n+1))$-bit keys and retain their comparison bits. Sentinels and original-position labels are included in the keys, so every comparison is deterministic, including on equal or malformed data. Reversing swaps and comparisons restores the records and zeros the comparison bits. A scan uses a fresh output word at each step and retains its old words until reversal. Modular fixed-width arithmetic gives a total Boolean function on every input; valid inputs fit in the chosen capacities. For predicates, accumulator widths are chosen for *all physical encodings*, not only valid tableaux. A sum of M unsigned w-bit row lengths uses $w + \lceil \log_2 M \rceil + 1$ bits, and differences are signed and widened. Alternatively, an overflow flag is retained and rejected. Thus wraparound cannot turn a malformed shape into an accepted partition or cover. All these widths remain $O(\log(n+1))$.

For columns, sort records (p_j, j) by row and then label. A segmented prefix count gives the rank of each label within its row. Sorting by original label places the resulting ranks at public output addresses. For the histogram, insert one sentinel for each row $r = 1, \ldots, M$ after that row's data records. Its segmented count is the row length; sort these sentinels by their public row labels for extraction. Check separately that all input rows lie in $[1, M]$. The row histogram must be nonincreasing. To test tableau validity, sort label records by their computed column and then row. Within each column, check that rows are consecutive starting at one and that labels increase. Within-row labels already increase by the definition of a row word. This proves necessity and sufficiency, including absence of holes. Partition validity is a fixed scan of row comparisons, nonnegativity, and the total sum. Cover validity checks that the padded rowwise differences are zero except for one $+1$, with both partitions valid and the correct total sizes.

Each algorithm is made into a clean function oracle by computing its intermediate history, copying the requested outputs by XOR, and reversing the history. Thus it works even when the target XOR register is nonzero. Malformed input is not an excuse to drop information: its computed predicate is copied, then the same computation is reversed. No lookup by a data-dependent row address is used.

For R_I, test all of the following: λ is a partition of $M-1$; the first $M-1$ row symbols encode a standard tableau whose endpoint is exactly λ; the final row symbol is zero; $1 \le i \le M$; and the attenuation flag and every W-work bit are zero. For R_J, test that $\lambda \vdash M-1$, all M row symbols encode a standard tableau Q whose endpoint μ covers λ, and the tag, flag, and every W-work bit are zero. Untested right paths and future digits are tensor-identity spectator factors in both projectors. Compute the conjunction in separate clean helper space, apply phase -1 precisely when it is false, and uncompute. All this is exact reversible Boolean logic. The helpers are *not* borrowed from possibly dirty W-work. □

One bookkeeping point is essential later: the column cache is built by a *clean* XOR oracle before applying representation matrices. We do not leave sorting history attached to the old tableau. After the tableau changes, the same XOR oracle evaluated on the new tableau erases the updated cache.

5 Elementary coherent branching coefficients

Let $x_0 > y_0 > x_1 > \cdots > y_{k-2} > x_{k-1}$ be the integer contents of addable and removable boxes of λ, respectively. The standard dimension-ratio identity gives, for the successor belonging to x_i,

$$p_i = \frac{d_{\lambda+\square_i}}{M d_\lambda} = \frac{\prod_{j=0}^{k-2} |x_i - y_j|}{\prod_{h\neq i} |x_i - x_h|}. \tag{5}$$

See also Lemma 2.4 of [5]. Normalization follows directly by taking the coefficient of $1/z$ in the partial fractions of $\prod_j (z - y_j)/\prod_i (z - x_i)$. A scan of padded row lengths identifies corners; a fixed stable compaction by sorting puts active corners first. There are k addable corners and $k-1$ removable corners, with

$$\frac{k(k-1)}{2} \le M - 1, \qquad L := x_0 - x_{k-1} \le 2M.$$

Indeed $k-1$ distinct positive row lengths require at least $1 + \cdots + (k-1)$ boxes. Thus all arrays may be padded to the public least power-of-two capacity $K = 2^{\lceil \log_2(\lceil \sqrt{2M} \rceil + 1) \rceil}$, so $K^2 = O(M)$.

Lemma 3 (Bounded products and precision)**.** *For $k > 1$, each probability in (5) has the factorization*

$$p_i = \prod_{j<i} \frac{y_j - x_i}{x_j - x_i} \prod_{j\ge i} \frac{x_i - y_j}{x_i - x_{j+1}}, \tag{6}$$

with every factor strictly between zero and one, and $p_i \ge L^{-2}$. For each i start with $N_i^{(0)} = 2^b$ and, for a factor a/d, update

$$N_i^{(t+1)} = \left\lfloor N_i^{(t)} a/d \right\rfloor.$$

Writing $N_i = N_i^{(k-1)}$, $T = \sum_i N_i$, and $\widehat{p}_i = N_i/T$, the following bounds hold whenever the stated precision makes $T > 0$:

$$0 \le p_i - 2^{-b} N_i < (k-1)2^{-b}, \tag{7}$$

$$\|p - \widehat{p}\|_1 \le 2k(k-1)2^{-b}, \tag{8}$$

$$\left\| \sum_i (\sqrt{p_i} - \sqrt{\widehat{p}_i}) |i\rangle \right\| \le \sqrt{2k(k-1)2^{-b}}. \tag{9}$$

In particular $b = O(\log(M+1) + \log(1/\eta))$ suffices for a clean coherent preparation of $\sum_i \sqrt{p_i}|i\rangle$ with error at most η and $M\,\mathrm{poly}(\log(n+1), \log(1/\eta))$ elementary gates and work bits. The case $k = 1$ is an exact deterministic state.

Proof. Regrouping denominators gives (6). On the left of i, use $y_j \ge x_{j+1} + 1$, and telescope all but the last positive integer numerator, giving a lower bound $1/(x_0 - x_i)$ when this side is nonempty. On the right, use $y_j \le x_j - 1$ and obtain $1/(x_i - x_{k-1})$ in the same way. An empty side contributes one. This proves $p_i \ge L^{-2}$. Each rounded update contracts the previous error and adds less than 2^{-b}, proving (7). If $\Delta = 1 - 2^{-b} \sum_i N_i$, then $0 \le \Delta \le k(k-1)2^{-b}$. Renormalization changes the dyadic vector by exactly Δ in ℓ^1, proving (8). The inequality $(\sqrt{a} - \sqrt{b})^2 \le |a - b|$ proves (9).

For example, take the public precision

$$b \ge \max\left\{ \left\lceil \log_2(2(K-1)(2M)^2) \right\rceil, 2\left\lceil \log_2(4K/\eta) \right\rceil \right\}.$$

The weight loss is at most $(K-1)2^{-b} \le 1/[2(2M)^2] \le p_i/2$, so every active N_i is positive. Weight-rounding state error is less than $\eta/2$. Inactive leaves are exactly zero. Throughout an update $N_i \le 2^b$ and its product with a numerator is at most $2^b L$; there are no exponentially large intermediate products. Fixed multiplication and division use polynomially many gates in $b+\log(M+1)$. Compute one product history, copy its final word, and reverse that history before moving to the next i. This uses $O(K^2)$ updates and $O(K)$ words of history at a time, up to polynomial word-arithmetic overhead.

Build the binary tree of the integer masses N_i. This uses the familiar conditional-mass preparation pattern [10], with the present integer weights and explicit rounding bounds. Each mass is at most $K2^b$. At a node with child masses A, B, use the real rotation of angle $\theta = \operatorname{atan2}(\sqrt{B}, \sqrt{A})$, controlled by the already prepared prefix. Zero children use exact identity or bit-flip endpoints; a zero parent uses identity. There are $K-1$ fixed nodes. Compute each angle to error at most $\eta/(8K)$ and synthesize its controlled rotation to operator error at most $\eta/(8K)$. Telescoping the gates gives error at most $\eta/4$, leaving slack in the claimed bound. Each node's prefix condition uses at most $\log_2 K$ bits: compute its equality flag in clean helper bits, apply the controlled rotation, and uncompute the flag. This adds only polynomial-logarithmic cost per node, with reusable helper space.

These angle computations need no oracle: evaluate $z = \sqrt{\min(A,B)/\max(A,B)} \in [0,1]$ by integer division and a fixed integer-square-root routine to enough dyadic places. For $z \le 1/2$ the alternating series for $\arctan z$ converges geometrically. For $z > 1/2$ use $\arctan z = \pi/4 - \arctan((1-z)/(1+z))$; the new argument is at most $1/3$. Machin's formula computes $\pi/4 = 4\arctan(1/5) - \arctan(1/239)$ to the same accuracy. Put $c = O(\log(K/\eta))$. Evaluate the $O(c)$ terms by fixed-point Horner evaluation, rounding each operation at $c + O(\log c)$ fractional bits; these guard bits bound the accumulated arithmetic error by $O(2^{-c})$. Exact rational powers are useful reference certificates but are not the claimed bounded-word circuit implementation. The square-root input can be evaluated to twice the desired output bits, using $|\sqrt{u} - \sqrt{v}| \le \sqrt{|u-v|}$. If $B > A$, subtract the small angle from $\pi/2$. Every step has polynomial bit complexity.

The $O(c)$ angle bits control $O(c)$ fixed dyadic rotations. Assign each bit-controlled rotation synthesis error $O(\eta/(Kc))$; their sum is $O(\eta/K)$ per tree node. Finite-gate synthesis has polynomial-logarithmic word length [8]. For classical compilation in a finite-precision bit model we use Theorem 4.5 of [9], rather than interpreting an arithmetic-operation count as a bit-complexity bound. The approximation here is of the actual controlled gate in matrix norm, not merely a one-qubit operation up to an uncontrolled global phase. For instance a controlled real rotation decomposes into two CNOTs and a half-angle rotation paired with its exact circuit adjoint; its irrelevant single-qubit synthesis phase then cancels. Prefix and angle-bit conjunctions are computed and uncomputed by exact Boolean circuits. Their cost is included in the polynomial-logarithmic factor. Compute, rotate, and uncompute the angle scratch. Tree masses, weight words, and contents are erased from the unchanged shape after preparation. The loop bounds use K, not the data-dependent k. Dummy factors are one, dummy leaves zero, and all invalid denominators are replaced by one in totalized arithmetic. A shape-validity guard may select a fixed default state on invalid shapes. All controlled maps therefore have a unitary extension, with no need to specialize or enumerate circuits over partitions. □

To obtain the actual added row rather than its corner ordinal, use a fixed scan of the K corner records: controlled on ordinal equality, XOR the selected row into the final blank tableau slot. A second fixed scan, controlled on equality with that unique row, erases the ordinal. This is reversible on the prepared support. On the complement it is still a composition of reversible XOR gates. The parameter table is then erased using the unchanged λ. Thus Lemma 3 implements the branching

column map

$$B : |P, 0\rangle \longmapsto \sum_{\mu \succ \lambda} \sqrt{d_\mu/(M d_\lambda)} |P \to \mu\rangle \tag{10}$$

with the same clean-error and cost bounds. A fixed-angle approximate circuit can leak into padded states; its entire leakage is included in the isometry error, not assumed to vanish exactly.

6 The selected transversal in near-linear cost

Write $s_j = (j, j+1)$ and $d = \mathrm{cont}(j+1) - \mathrm{cont}(j)$ in a valid tableau. Young's orthogonal rule is

$$R_\mu(s_j)|P\rangle = d^{-1}|P\rangle + \sqrt{1 - d^{-2}}|s_j P\rangle.$$

When $|d| = 1$ the second term is absent and the action is the exact sign d. Otherwise $s_j P$ is standard and its axial distance is $-d$.

Lemma 4 (Cached generators)**.** *With a correct column cache, one public-index Young generator, also controlled on a public comparison with the tag, can be implemented to operator error ζ on the valid-tableau/cache subspace in* $\mathrm{poly}(\log(n+1), \log(1/\zeta))$ *gates and work bits. The ideal operation preserves that subspace and updates the column cache correctly.*

Proof. Only the two records (p_j, c_j) and (p_{j+1}, c_{j+1}) are accessed. For $|d| > 1$, compute $f = [d < 0]$ and swap both records if $f = 1$, thereby canonicalizing the pair to a positive distance $D = |d|$. Retain f; this step must not discard which member of the pair was present. On f apply

$$A_D = \begin{pmatrix} 1/D & \sqrt{1 - D^{-2}} \\ \sqrt{1 - D^{-2}} & -1/D \end{pmatrix} = R_y(2 \arccos(1/D)) Z.$$

Compute and erase D and the angle while the canonical records are unchanged. Undo the record swap controlled on the now-superposed f. Finally XOR $[d_{\mathrm{final}} < 0]$ into f, erasing it on both branches. The $|d| = 1$ branch applies the exact sign; its guard is invariant and can be erased. On invalid inputs use safe total arithmetic and a reversible extension of this gate sequence, without making a promise of cleanup.

The swapped column cache is correct because $s_j P$ interchanges the boxes containing $j, j+1$; all other box coordinates are unchanged. If $\theta = \arccos(1/D)$, then $\|A_D(\theta) - A_D(\widetilde{\theta})\| = 2|\sin((\theta - \widetilde{\theta})/2)| \le |\theta - \widetilde{\theta}|$. Compute θ by the same bounded-ratio square-root/arctangent routine as above, or via $\pi/2 - \arctan(1/\sqrt{D^2 - 1})$. The latter ratio is at most $1/\sqrt{3}$; a further range reduction gives a geometrically convergent series. Word precision $O(\log(n+1) + \log(1/\zeta))$ suffices. Arithmetic is uncomputed and controlled dyadic rotations are synthesized with the claimed polynomial-logarithmic cost. For a c-bit angle word, assign each of its $O(c)$ controlled constant rotations error $O(\zeta/c)$, using the same phase-aware construction as above. The axial-distance sectors are orthogonal controls, so the error is the maximum sector error, not a sum over tableaux. Only the subsequent $M - 1$ generator applications are summed; $\zeta = O(\eta/M)$ suffices for the selected transversal. □

Build the clean column cache of the complete M-box tableau, then, in chronological order $j = M-1, M-2, \ldots, 1$, apply this generator controlled on $[i \le j]$. Since $t_i = s_i s_{i+1} \cdots s_{M-1}$, this order implements

$$V : |i, Q\rangle \longmapsto |i\rangle R_\mu(t_i)|Q\rangle.$$

The tag zero case can act as identity. The cache is updated at each swap, and is erased by the clean column XOR oracle evaluated on the final tableau, not by reversing stale pre-generator history. With per-generator error $O(\eta/M)$, the whole selected operation has clean error $O(\eta)$ and cost $M\,\mathrm{poly}(\log(n+1), \log(1/\eta))$. This error statement follows by telescoping on ideal intermediate valid-cache subspaces; it does not require approximate intermediate states to satisfy exact validity.

7 Column erasure and full-unitary amplification

Let H_M be the specific tag unitary defined by a fixed binary preparation tree, with $H_M|0\rangle = M^{-1/2}\sum_{i=1}^{M}|i\rangle$. The tree has $O(M)$ leaves; approximating each of its fixed rotations gives a whole-unitary operator-norm approximation of H_M, and hence of its adjoint, to error η in $M\,\mathrm{poly}(\log(n+1), \log(1/\eta))$ gates. No stronger cost bound for this elementary uniform state is needed. Define the ideal unitary

$$W_0 = (H_M^\dagger \otimes \mathrm{I})VB.$$

Projecting onto tag zero and all work zero gives the exact block identity

$$J^\dagger W_0 I = F/\sqrt{M}. \tag{11}$$

Projection here specifies a matrix block; it is not a measurement.

The bra of the uniform tag state contributes the same factor $1/\sqrt{M}$ to every retained tag. The remaining coefficients are exactly those in (3). Thus the good block is a scalar multiple of a unitary on *every* input, not merely a state with a large overlap for one chosen column. This input-independent normalization is what oblivious amplification needs.

Choose public values

$$r_M = \lceil \sqrt{M} \rceil, \qquad \theta_M = \frac{\pi}{4r_M + 2}, \qquad a_M = \sqrt{M}\sin\theta_M. \tag{12}$$

Since $a_M \le \pi/4 < 1$, a one-qubit flag rotation with zero amplitude a_M is legal. Apply it in addition to W_0 and call the result W. The target tests flag zero, giving

$$J^\dagger W I = \sin\theta_M F. \tag{13}$$

Computing and synthesizing this public angle has polynomial-logarithmic cost. Special-case $M = 1$ as a trivial induction if desired.

Lemma 5 (Two-promise OAA and its error)**.** *Suppose I, J are isometries, F is unitary between their equal-dimensional domains, and a unitary W obeys $J^\dagger W I = \sin\theta F$ with $\theta = \pi/(4r+2)$ for an integer $r \ge 1$. Then*

$$(-W R_I W^\dagger R_J)^r W I = JF. \tag{14}$$

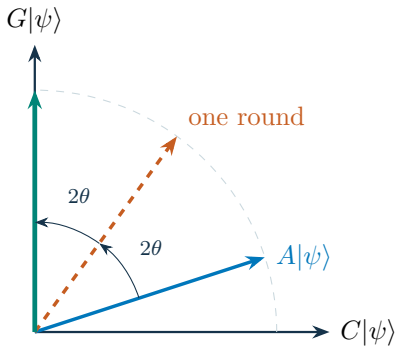
If $\widetilde{W}$ is unitary and $\|(\widetilde{W} - W)I\| \le \eta$, keeping the exact promise reflections gives clean output error at most $(4r+1)\eta$.

Proof. Put $A = WI$, $G = JF$, $s = \sin\theta$, $c = \cos\theta$, and $C = (A - sG)/c$. The block identity implies $G^\dagger C = 0$ and $C^\dagger C = \mathrm{I}$, with $A = sG + cC$. In particular $J^\dagger C = 0$. On each corresponding two-dimensional plane, $-WR_IW^\dagger R_J = -(2AA^\dagger - \mathrm{I})(2JJ^\dagger - \mathrm{I})$ sends

$$G \mapsto \cos(2\theta)G - \sin(2\theta)C, \qquad C \mapsto \sin(2\theta)G + \cos(2\theta)C.$$

Its rth power applied to A is $\sin((2r+1)\theta)G + \cos((2r+1)\theta)C = G$. The minus sign in (14) is included; it avoids an unwanted phase convention. This is the usual oblivious-amplification geometry [7], written for distinct complete source and target promises.

For the error bound, set $\widetilde{A} = \widetilde{W}I$. Then $\|\widetilde{A}\widetilde{A}^\dagger - AA^\dagger\| \le 2\eta$, so the two image reflections differ by at most 4η. Telescoping the r rounds and the initial isometry gives $(4r+1)\eta$. Only the action of W on the whole source promise must be approximated; no bound on its arbitrary unitary completion is needed. The implemented inverse is the exact adjoint circuit $\widetilde{W}^\dagger$, not a separately synthesized inverse. If the two reflections themselves had global errors ρ_I, ρ_J, one would add $r(\rho_I + \rho_J)$. Here the Boolean promise reflections are exact. □



Exact geometry at $M = 4$

$r = 2, \quad \theta = \pi/10$

Start: $\sin\theta\, G + \cos\theta\, C$

After one round: $\sin 3\theta\, G + \cos 3\theta\, C$

After two rounds: G

Every source basis vector has its own plane. The same circuit acts coherently on all of them.

Figure 3: Amplification is rotation, not repeated measurement. In the ideal construction, G is the desired clean output and C is an orthogonal complement. Attenuation chooses the starting angle so an integer number of rounds reaches G exactly. Dashed and solid arrows distinguish intermediate and initial/final states.

One induction level, in circuit order. The construction can be read as the following fixed schedule.

1. Prepare the successor tableau with B, retaining the input tag.
2. Build its column cache, apply the $M-1$ selected adjacent generators, and erase the updated cache. This is V.
3. Apply $H_M^\dagger$ to the tag and the public attenuation rotation to the flag. Together, these operations are W.
4. Repeat r_M times: R_J, $W^\dagger$, R_I, W, and the overall factor $-\mathrm{I}$, in that chronological order.
5. Update the outer shape and right tableau as described next.

Every inverse in step 4 is the literal adjoint of the chosen finite circuit. None of these steps selects a measured outcome.

Lemmas 2–4, the tag preparation, and the flag rotation together construct $\widetilde{W}$ with source-isometry error η at cost $M\,\mathrm{poly}(\log(n+1), \log(1/\eta))$. Split this budget among branching preparation, the selected generators, tag preparation, and the flag. At fixed λ the source tableau is a spectator during branching preparation, so its state-error bound is an operator-isometry bound. Direct sum over λ

takes a maximum of these errors, not a sum over the number of partitions. The public capacities and number of amplification rounds are the same for every coherent λ sector.

There are $2r_M + 1$ calls to $\widetilde{W}$ or its adjoint and $2r_M$ promise reflections. Thus a complete clean induction step costs

$$M^{3/2}\,\mathrm{poly}(\log(n+1), \log(1/\eta)) \tag{15}$$

gates and $M\,\mathrm{poly}(\log(n+1), \log(1/\eta))$ work bits. No branch is postselected and no shape-dependent angle table is supplied for free.

8 Assembling the full transform

At stage M the previous Fourier registers have shape λ and tableaux $P, R \in \mathrm{SYT}(\lambda)$. Apply the amplified F_λ to (i_M, P), leaving R unchanged. On the ideal output, compute the new shape μ from Q, and compute the unique row a where $\mu - \lambda$ equals one. Append a to R to form $R \to \mu$.

This boundary map is clean and reversible in $\widetilde{O}(M)$ cost. For example, keep the newly computed μ, transfer a to the blank last right-path slot, and erase temporary difference work while λ is still available. Then XOR the reconstructible old shape $\mu - \square_a$ into the old λ register. It becomes zero. Swap the new and old shape registers, retaining only μ. The former new-shape register is now zero. All row scans, histograms, and compute-copy-reverse operations have fixed schedules. On malformed inputs these same XOR operations still define a reversible circuit; residuals need not vanish. On approximate inputs their norm is charged to the preceding error.

To verify (2) inductively, write $g = t_i^{(M)} h$ with $h \in S_{M-1}$. The old coefficient times the induction coefficient is

$$\sqrt{\frac{d_\lambda}{(M-1)!}}\,\langle P|R_\lambda(h)|R\rangle\,\sqrt{\frac{d_\mu}{M d_\lambda}}\,\langle Q|R_\mu(t_i)|P \to \mu\rangle.$$

Sum over P and use the adapted restriction of R_μ to the λ sector. This is $\sqrt{d_\mu/M!}\langle Q|R_\mu(t_i h)|R \to \mu\rangle$, exactly the next Fourier coefficient. Starting from the empty tableau proves the whole recurrence, including the right-path update.

Proof of Theorem 1. Use the public per-column error budget

$$\eta = \frac{\epsilon}{2n(4\lceil\sqrt{n}\rceil + 1)}. \tag{16}$$

By Lemma 5, the sum of clean induction errors is at most $n(4\lceil\sqrt{n}\rceil + 1)\eta = \epsilon/2$. Exact boundary maps do not increase it. This hybrid argument evaluates each individual error on the ideal valid input to that stage; actual accumulated leakage is carried unitarily and requires no intermediate projection or measurement. More explicitly, let U_j be ideal stages including their exact boundaries, let $\widetilde{U}_j$ be their implementations on the same physical registers, and let E_{in} embed a valid clean input. Then

$$\|(\widetilde{U}_m \cdots \widetilde{U}_1 - U_m \cdots U_1)E_{\mathrm{in}}\| \le \sum_{j=1}^{m} \|(\widetilde{U}_j - U_j)U_{j-1} \cdots U_1 E_{\mathrm{in}}\|. \tag{17}$$

The omitted approximate suffixes are unitary and cannot enlarge norms. Each ideal prefix has clean work, so each summand satisfies the local promise. Actual work need not be reset between stages. The separate predicate helpers remain exactly clean and are untouched by W. All word precisions

are $O(\log(n+1)+\log(1/\epsilon))$, up to guard bits of the same asymptotic size. Summing (15) gives

$$\sum_{M=1}^{n} O(M^{3/2})\,\mathrm{poly}(\log(n+1),\log(1/\epsilon)) = n^{5/2}\,\mathrm{poly}(\log(n+1),\log(1/\epsilon)).$$

Scratch space is reused across calls and stages, and all persistent transversal, shape, and tableau words occupy $O(n\log(n+1))$ bits. This proves the live-space bound. Reversible multiplication, division, sorting, and finite-precision series have fixed polynomial-time Boolean circuits. Fixed dyadic rotations are synthesized uniformly in polynomial time in their precision in the bit model [9]. Hence the circuit family is uniform. Explicit conservative logarithmic exponents and the classical compilation bound are given in Section 9. Finally, clean isometry error at most $\epsilon/2$ implies channel diamond error at most ϵ, also after tracing out work. Indeed, for the ideal clean isometry V_n and the implemented $\widetilde{V}_n$, the associated channels satisfy

$$\|\widetilde{\mathcal{F}}_n - \mathcal{F}_n\|_\diamond \le 2\|\widetilde{V}_n - V_n\| \le \epsilon.$$

Tensoring by a reference identity preserves the operator-norm bound, and the trace norm is contractive under discarding work. □

8.1 Input-encoding cost

A one-line permutation $g(1),\dots,g(n)$ can be converted within the same dominant bound without relying on a fast encoding theorem. At the current size M, move the last image $i_M = g(M)$ into a blank digit slot, leaving the last image slot zero. On each of the remaining images apply $(t_{i_M}^{(M)})^{-1}$: the value i_M maps to M, larger values decrease by one, and smaller values are unchanged. This is a genuine permutation of $[1,M]$, not a noninvertible conditional decrement. On valid inputs the remaining images never equal i_M, so they now form a permutation of $[1,M-1]$. Continue downwards. Reversal reconstructs the original permutation from the saved digits, without retaining earlier arrays. Public scans make $O(n^2)$ controlled word permutations, each implemented by reversible comparisons and arithmetic in polynomial word cost and space. For example, compute its piecewise bijection into a temporary word, exchange the words, and erase the temporary by its inverse piecewise bijection. Invalid digit values can select identity and out-of-range image values can be fixed. Thus the full conversion is a reversible unitary extension using $O(n\,\mathrm{poly}(\log(n+1)))$ space, and its $O(n^2\,\mathrm{poly}(\log(n+1)))$ gate cost is lower-order than the QFT cost.

9 Explicit precision and resource audit

The following refinement makes the suppressed costs checkable. Define

$$\Lambda = 2 + \lceil \log_2(n+1) \rceil + \lceil \log_2(1/\epsilon) \rceil,$$

and use η from (16). These are sufficient bounds, not optimized logarithmic exponents or predictions of practical circuit size.

Proposition 6 (Explicit resource bounds)**.** *The construction can be chosen to use*

$$O(n^{5/2}\Lambda^5) \text{ elementary Clifford+}T \text{ gates}, \qquad O(n\Lambda^3) \text{ live logical qubits}.$$

A deterministic classical algorithm emits the circuit in $n^{5/2}\,\mathrm{poly}(\Lambda)$ bit operations. Shape-dependent angle arithmetic is performed coherently inside the circuit and is included in the gate bound, not charged only to this public classical compilation.

Proof. At level $M \geq 2$, take the public corner capacity K above and set

$$b = \max\left\{\left\lceil \log_2(2(K-1)(2M)^2)\right\rceil, 2\left\lceil \log_2(16K/\eta)\right\rceil\right\},$$
$$c = \left\lceil \log_2 \frac{256(M+K+1)}{\eta}\right\rceil, \qquad q = c + \lceil \log_2(c+2)\rceil + 12.$$

Every parameter is $O(\Lambda)$. The positivity estimate of Lemma 3 applies, and the weight-state error is at most $\eta/8$.

Here is a fully bounded-word angle algorithm. For positive integer masses A, B, find the largest integer v with

$$v^2 \max(A,B) \leq 2^{2q} \min(A,B)$$

by $q+1$ fixed bit trials. Thus $r = v2^{-q}$ approximates the desired square root from below within 2^{-q}. When $r > 1/2$, set $w = \mathrm{floor}_q((1-r)/(1+r))$ and later subtract $\arctan w$ from $\pi/4$; otherwise set $w = r$. In either case $0 \leq w \leq 1/2$. Evaluate

$$\arctan w \approx w \sum_{j=0}^{q+1} \frac{(-1)^j (w^2)^j}{2j+1}$$

by signed Horner evaluation, rounding each multiplication and each public coefficient to q fractional bits. Two additional integer/sign bits suffice for the accumulators. Compute the public $\pi/4$ by the same routine and Machin's formula, with the rational arguments rounded down; double it for $\pi/2$. Complement the angle when $B > A$.

For completeness, put $u = 2^{-q}$. The rounded Horner multiplier is at most $1/4$, the exact intermediate polynomials have magnitude at most $4/3$, and each coefficient, product, and multiplier rounding contributes at most a constant times u. The resulting error recurrence is bounded by $e_j \leq e_{j+1}/4 + 4u$. The series remainder is below u. Root rounding contributes at most u because $|\arctan'| \leq 1$; range reduction adds at most u, using the exact identity at the rounded r. Machin's coefficients have absolute sum five; complementing adds at most two copies of the public constant error. Summing these estimates is bounded conservatively by

$$64(q+1)2^{-q} < 2^{-c}.$$

Zero parents select identity, and one-sided masses select exact endpoint gates. No ratio lower bound or exponentially large exact rational powers are needed. Public tag angles use the same routine.

For attenuation, a concrete sufficient guard rule is $h = c + \lceil \log_2(M+1)\rceil + 12$. Compute $\sqrt{M}$, θ_M, and the sine at the approximated angle to absolute error at most 2^{-h}, then round their product at the same precision. Integer root trials, Machin's formula, and $O(h)$ sine-series terms at $h + O(\log(h+1))$ working bits suffice; the sine argument is below one. The propagated product error is at most $(2\sqrt{M}+4)2^{-h} < 2^{-c-8}$, using $|\sin'| \leq 1$. Clip the approximate a_M to $[0, 4/5]$. This does not increase its distance from the exact $a_M \leq \pi/4$, and $|(\arccos)'| \leq 5/3$ throughout that interval. Evaluate

$$\arccos a = 2\arctan\sqrt{\frac{1-a}{1+a}}$$

by the bounded-ratio routine at $c+4$ target bits. Its approximation error, including the factor two, and the propagated error in a sum to less than 2^{-c}. All constants are public and computed in polynomial bit time. This completes the precision accounting for the flag rotation.

Let $D = 2^{\lceil \log_2(M+1) \rceil}$. There are at most $N = K + M + D - 2$ angle calls: $K - 1$ branching nodes, $M - 1$ generators, $D - 1$ tag nodes, and one attenuation. Thus their summed arithmetic error is below $\eta/64$. There are at most

$$S = (K + M - 2)(q + 2) + D$$

public rotation slots after expanding each computed angle into its bits and allowing for endpoints. Assign each slot synthesis error $\delta = \eta/(16S)$. Its total contribution is at most $\eta/16$. Together with weight rounding this is strictly less than η, with slack for the public-constant evaluation. Controlled rotations use the phase-cancelling exact-adjoint construction already described, synthesizing each half rotation to error at most $\delta/2$. An uncontrolled rotation can use that construction with one clean auxiliary temporarily set to one and restored. Overall minus signs can be realized exactly by the Clifford identity $XZXZ = -\mathrm{I}$ at constant cost.

On $O(\Lambda)$-bit words, addition and comparisons cost $O(\Lambda)$ Boolean gates, and schoolbook multiplication and restoring division cost $O(\Lambda^2)$. The $O(\Lambda)$ root trials and Horner steps therefore cost $O(\Lambda^3)$ per angle. Retaining a Boolean computation history uses at most a constant number of fresh bits per gate; compute–copy–reverse only changes constants. Sorting $O(M)$ records requires $O(M \log^2 M)$ comparisons of $O(\log n)$-bit records, hence $O(M\Lambda^3)$ gates and retained history. All $K(K-1) = O(M)$ weight updates cost $O(M\Lambda^2)$. Even retaining every weight-update history simultaneously would stay within this bound.

Finite synthesis can use $O(\log^4(1/\delta)) = O(\Lambda^4)$ gates per public rotation. For a bit-model uniform version, use [9]; its single-qubit construction has five emitted recursive pieces per precision improvement $\varepsilon_\ell = O(\varepsilon_{\ell-1}^{3/2})$, giving word-length exponent $\log_{3/2} 5 < 4$, and polynomial bit compilation time. Phase-normalized single-qubit generators may be replaced by their physical Clifford+T representatives inside the cancelling sandwich. The $O(\Lambda)$ rotation bits per angle therefore give $O(\Lambda^5)$ gates per angle. Synthesis words act on constant-size registers and do not require a new history qubit per emitted quantum gate.

It follows that one column uses $O(M\Lambda^5)$ gates and $S_W = O(M\Lambda^3)$ work bits. A complete promise reflection tests S_W work bits by a linear conjunction, in addition to the $O(M\Lambda^3)$ tableau predicates. Its separate clean helpers also occupy $O(M\Lambda^3)$ bits. The helpers are not part of W and do not recursively need their own validity test. Amplification repeats gates, not physical registers; its $O(\sqrt{M})$ repetitions give the claimed sum over M. Exact outer boundaries cost $O(M\Lambda^3)$ per stage. The hybrid proof permits reuse even when actual scratch has small leakage.

All public angles are polynomial-bit computable, the finite synthesis algorithm has polynomial bit cost, and the fixed Boolean schedules can be emitted in polynomial time per word operation. Including wire addresses and writing the complete output gives $n^{5/2}\,\mathrm{poly}(\Lambda)$ classical bit time. This statement does not claim that the original synthesis arithmetic-operation exponent is its bit-complexity exponent. □

10 Resource ledger and interpretation

Table 3 collects the costs proved above. Every entry includes reversible parameter computation and cleanup. The explicit bound in Proposition 6 uses conservative logarithmic exponents; the table suppresses them to expose the dependence on M and n.

Two different meanings of classical cost. If a rotation angle depends on a quantum shape, computing it is part of the quantum algorithm: reversible Boolean gates must calculate, use, and

Operation at size M	Elementary gates	Live work qubits
Corners and $K^2 = O(M)$ bounded products	$\widetilde{O}(M)$	$\widetilde{O}(M)$
Branching tree and ordinal/row conversion	$\widetilde{O}(\sqrt{M})$	$\widetilde{O}(M)$
Cache construction and cleanup	$\widetilde{O}(M)$	$\widetilde{O}(M)$
Selected $M-1$ Young generators	$\widetilde{O}(M)$	$\widetilde{O}(M)$
Tag preparation and attenuation	$\widetilde{O}(M)$	$\widetilde{O}(M)$
Each complete promise reflection	$\widetilde{O}(M)$	$\widetilde{O}(M)$
Amplified induction	$\widetilde{O}(M^{3/2})$	$\widetilde{O}(M)$
Outer boundary update	$\widetilde{O}(M)$	$\widetilde{O}(M)$
Complete S_n QFT	$\widetilde{O}(n^{5/2})$	$\widetilde{O}(n)$

Table 3: Gates accumulate; workspace is reused. The branching-tree row excludes computing its weights, which is charged in the first row. There is no shape-dependent classical oracle hidden in the ledger.

erase that angle coherently. In contrast, the classical compiler generates a fixed circuit from the public size and precision parameters. Its $n^{5/2}\,\mathrm{poly}(\Lambda)$ bit cost includes writing the circuit and synthesizing its public constant rotations. It never enumerates the partitions or chooses a circuit after observing a quantum register. Both costs are accounted for; one cannot replace the other.

What has, and has not, improved. Relative to the cubic full-QFT theorem stated in [5], the exponent of n decreases by $1/2$. This is not a priority determination or a proof of practical advantage. The near-linear space bound is established for the present construction, but we do not claim a strict separation from all previous implementations: the proof-level workspace discussion in Section 6 and Appendix A.2.1 of [5] is tighter than its stated Theorem 1.1 bound suggests. Nor does this construction prove that $5/2$ is optimal. It bypasses the local interlacing A-matrix decomposition; it does not improve that matrix's exact Givens count.

11 Validation and reproducibility

The theorem is analytic. Numerical checks test conventions, reversibility, and implementation contracts; they do not establish the asymptotic bound. The accompanying repository separates three levels of evidence.

Independent small-instance identities. Dense reference calculations check the induction matrices, their two-promise amplification completions, and the full small-group Fourier transform. Separate contract checks exercise cached-generator signs, transversal order, promise padding, and the outer right-tableau update. Generic complex-matrix tests change off-promise unitary completions and reuse dirty scratch across stages, checking precisely the error mechanisms in Lemma 5 and (17).

Arithmetic certificates. Exact-rational and directed-bound references check branching products, integer rounding, and angle enclosures. The integer-only guarded-Horner reference implements the public precision schedule of Proposition 6. It is a mathematical and computational reference, not yet a reversible arbitrary-precision compiler.

Finite-gate prototype. A separate unoptimized compiler emits complete small Clifford+T programs, including shape-dependent arithmetic, promise reflections, and cleanup. Table 4 reports the saved nine-bit runs. All six source columns of the $M = 3$ induction step, across both shapes, were checked. For the full S_3 circuit, every input column was compared with Fourier coefficients computed directly from Young representation matrices, independently of the induction recurrence.

Circuit	Finite gates	Logical qubits	Clean operator error
Three-box induction	17,138,782	76,009	0.00505832
Full S_3 QFT	29,545,004	76,019	0.00507360

Table 4: Implementation evidence at nine fractional weight and angle bits. These are logical, non-error-corrected counts, not practical resource estimates. Overall-sign annotations are excluded from the gate count: two for induction and four for the full transform.

The full-S_3 program uses one shared work pool without projection or reset. Its measured leakage norm is 0.00507181; a complex-superposition test of the circuit followed by its literal adjoint has residual norm 3.23×10^{-10}. The pruning contribution to the reported operator comparison is bounded by 2.44×10^{-12}, but floating-point roundoff is not interval-certified. Two runs produced identical saved reports.

There is an important gap between this prototype and the theorem's precision prescription. The prototype uses rounded Taylor powers and host floating-point public constants, rather than the full guarded-Horner and public-constant scheme proved above. One-line permutation conversion is also not compiled. These checks therefore do not certify an arbitrary-precision implementation. Larger resource-only compilations must likewise not be interpreted as full-transform numerical validation.

Reproducibility materials. The accompanying source repository is `https://github.com/natestemen/fourier`. The saved gate reports are `data/direct_induction_gate_M3_b9_c9.json` and `data/full_qft_gate_S3_b9_c9.json`. The reference precision schedule is `experiments/direct_qft_precision_schedule.py`. The source-attribution audit records the exact inspected passages and access limitations in `data/paper_source_audit.json`; internal proof-review findings and responses are recorded separately. The full test suite can be run with `pytest` in the repository's configured Python environment. Reproducing the expensive gate simulations is a separate task from running those tests.

12 Discussion and next steps

The central simplification is to optimize the induction transform as a whole. Near-linear column preparation already contains its matrix coefficients; erasing the transversal tag exposes a uniformly normalized unitary block. Complete-promise amplification then pays a square-root overhead, while clean boundaries make the resulting steps composable. This yields a subcubic full-QFT construction with explicit coherent arithmetic and finite-gate accounting.

Three next steps would most directly strengthen the result. First, the guarded precision prescription should be compiled reversibly and checked with certified numerical error bounds. Second, arithmetic histories and synthesis constants should be reduced before making any practical resource comparison. Third, an independent human audit should check the representation conventions, full promises, and bit-uniform resource proof. The targeted literature check here is not an exhaustive priority search.

Lowering the exponent below 5/2 would require an additional idea. Within this construction, the two visible costs are the near-linear column/reflection work and the $\sqrt{M}$ amplification overhead. Reducing either cost, or sharing more work across levels, is a possible direction. None of these possibilities is ruled out by this paper.

Development and disclosure. This research draft was developed with AI assistance in derivation, coding, literature checks, manuscript preparation, and internal adversarial review. Such review is neither independent human peer review nor formal verification. No external review, publication acceptance, or exhaustive novelty certification is asserted.

References

[1] Robert Beals. *Quantum computation of Fourier transforms over symmetric groups*. Proceedings of STOC, 48–53 (1997). https://doi.org/10.1145/258533.258548.

[2] Cristopher Moore, Daniel Rockmore, and Alexander Russell. *Generic quantum Fourier transforms*. ACM Transactions on Algorithms 2(4), 707–723 (2006). https://arxiv.org/abs/quant-ph/0304064.

[3] Yasuhito Kawano and Hiroshi Sekigawa. *Quantum Fourier transform over symmetric groups*. Proceedings of ISSAC, 227–234 (2013). https://doi.org/10.1145/2465506.2465940.

[4] Yasuhito Kawano and Hiroshi Sekigawa. *Quantum Fourier transform over symmetric groups—improved result*. Journal of Symbolic Computation 75, 219–243 (2016). https://doi.org/10.1016/j.jsc.2015.11.016.

[5] Carli Bruinsma, Dmitry Grinko, and Maris Ozols. *Quantum Fourier transform for the symmetric group*. arXiv:2608.28569v2 (2026). https://arxiv.org/abs/2608.28569v2.

[6] Carli Bruinsma, Pietro M. Posta, Joppe Stokvis, Dmitry Grinko, and Maris Ozols. *Quantum Fourier transform toolbox*. arXiv:2608.28573v2 (2026). https://arxiv.org/abs/2608.28573v2.

[7] Dominic W. Berry, Andrew M. Childs, Richard Cleve, Robin Kothari, and Rolando D. Somma. *Exponential improvement in precision for simulating sparse Hamiltonians*. Proceedings of STOC, 283–292 (2014). https://arxiv.org/abs/1312.1414.

[8] Christopher M. Dawson and Michael A. Nielsen. *The Solovay–Kitaev algorithm*. Quantum Information & Computation 6(1), 81–95 (2006). https://arxiv.org/abs/quant-ph/0505030.

[9] Dieter van Melkebeek and Thomas Watson. *Time-Space Efficient Simulations of Quantum Computations*. Theory of Computing 8(1), 1–51 (2012). https://doi.org/10.4086/toc.2012.v008a001.

[10] Lov Grover and Terry Rudolph. *Creating superpositions that correspond to efficiently integrable probability distributions*. arXiv:quant-ph/0208112 (2002). https://arxiv.org/abs/quant-ph/0208112.

[11] Dikshant Rathore and Leo Zhou. *An efficient Hamiltonian-based quantum algorithm for characters of the symmetric group*. arXiv:2610.05752v1 (2026). https://arxiv.org/abs/2610.05752.

[12] Kagwe A. Muchane. *The Quantum Character Transform*. arXiv:2610.16827v1 (2026). https://arxiv.org/abs/2610.16827.